

ANOVA: ANOTHER EXPLANATION

- translates data into two diff estimates of variance

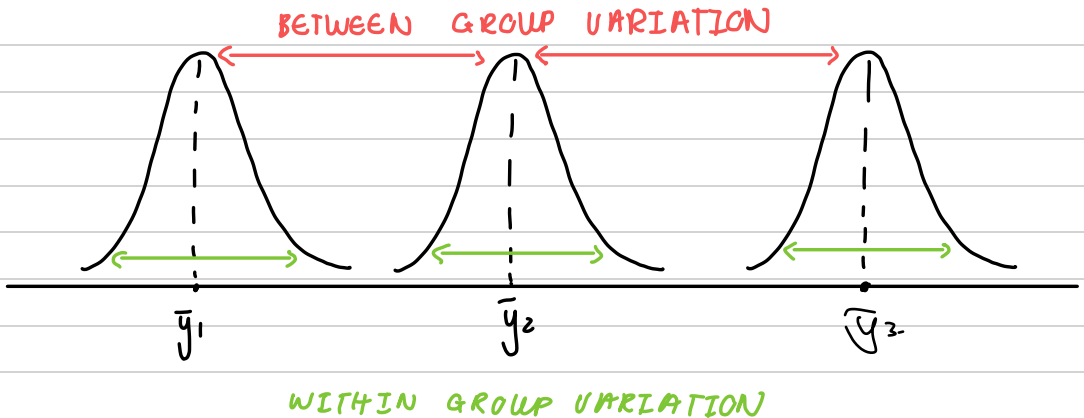
↳ does our treatment introduce variation into our data?

- if null hypo is TRUE, then treatment variation could be fully explained by random nature of data (residual variation)

↳ treatment did not introduce variation

VISUAL:

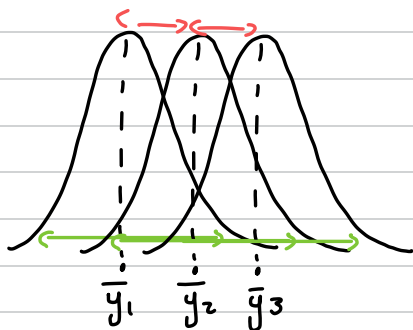
say H_0 is FALSE: between $\gg$ within



- treatment did introduce variation $\therefore$ had an effect

- treatment did have an effect $\Rightarrow$ at least one mean is different (in this case: all three) $F^* > 1$

• H_0 is TRUE: between $\approx$ within



$\rightarrow$ variation due to treatment is probably just due to sampling error / random variation in data

$\therefore$ treatment did not have an effect; means are equal ($\mu_1 = \mu_2 = \mu_3$)

$\rightarrow$ could also say: sample means are about as far apart as you'd expect if pop means were equal

$\rightarrow F^* \approx 1$